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# Comparative Evaluation of Automated User Acceptance Testing Tool for Web Based Application

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***Abstract*— User acceptance testing is the last phase of the software testing process. During user acceptance testing process, actual software users test the software to make sure it can handle required tasks in real-world scenarios, according to specifications. The purpose of this paper is to compare three automated user acceptance testing tool for web application. The evaluation is based on the usability criteria of the tools. The three tools evaluated are Acceptance Test-Driven Development. Three proposed tools are TestComplete, Selenium Webdriver and Watir Webdriver. These three tools undergo usability criteria testing that include efficiency, effectiveness, satisfaction and error. From the evaluation, Watir Webdriver has been proven as the best testing tool based on usability criteria of user acceptance testing.**

**Keywords — Acceptance Test-Driven Development, TestComplete, Selenium, Watir.**

## I. INTRODUCTION

User acceptance testing (UAT) is a process of verifying a solution works for user. It is not about ensuring software does not crash and meets requirements but its function is to ensure that solution will work for user. The purpose of UAT is to gather input from actual system users, those who have experience with the business processes and will be using the system to complete related tasks [7]. According to Jureczko and Mlynarki (2001), web application has to be tested in customer-like environment. The main reason for conducting UAT is to find out what the system can do before implementation. Since the test result tends to be taken more seriously a high user acceptance is important [1]. Besides, testing is the major component of any software engineering process meant to produce high quality application [2]. In short, keeping UAT goals at task and requirement level allows increase in software robustness and usability.

Nowadays, a wide range of open source and commercial existing automated tools are available. With the results of reviewing, there are three open sources automated testing tools had been selected to be reviewed in this paper. These three tools are Selenium Webdriver, TestComplete and Watir. Selenium Webdriver is the future direction of the project and the newest addition to Selenium toolkit as it provides all sorts of awesome features, including more cohesive and objects oriented API as well as an answer to limitation of the old implementation [9]. It supports the WebDriver API and underlying technology. Watir is an open-source family of Ruby libraries that supports multiple browsers on different platforms that allows writing tests that are easy to read and maintain in a simple and flexible manner. Watir and Selenium can support the automated test scenario for web based application [3]. TestComplete is an automated testing tool that creates, manages and runs tests for any windows, web or rich client software. It makes it easy for anyone to create automated tests with low price, powerful features and impressive support resources. TestComplete has easy to use UI, efficiency playback and will be the best to use for applications with lesser security needs [6]. So, a special method needs to be implemented as a technique in process of comparing these three tools.

The paper is organized as follows: in the next section, the background of methodology of this study will be described. Section 3 illustrates the evaluation results and discusses the findings from the results. Finally, section 4 describes the conclusion for the study.

## II. Methodology

Acceptance Test Driven Development (ATDD) method is a method that focuses on capturing requirements in acceptance tests and uses them to drive the development. ATDD is a software development technique that combines requirement specification with automatic executable tests of these requirements [4]. This work adopted ATDD design the test cases. The process of ATDD are create user story, capture acceptance criteria, convert scenarios which is create falling acceptance test and make test pass, and lastly verifiable working software. A user story, in context of ATDD, is a brief description of what exactly the upcoming acceptance test is going to test. The purpose of a user story is to catch the essential meaning of an acceptance test and act as a limit for that [8]. So, these four main steps are used to evaluate the three automated testing tools.

This research adopted Jureczko and Mlynarski [5] tool evaluation process in evaluating the tools. Figure 1 illustrates the process. As shown in Figure 1, the evaluation process is divided into three major steps; (i) tool implementation, (ii) answering questions and (iii) comparing tool.

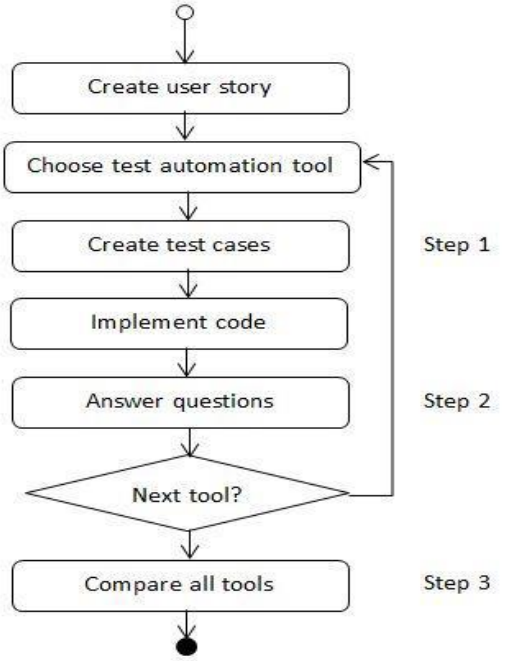


Fig. 1. Tool Evaluation Process

### *A. Comparative Evaluation Criteria*

The main criteria, usability as it comprises five other criteria which are learnability, efficiency, memorability, errors and satisfaction. Three criteria states by other researcher also seem important for evaluation process. They are effectiveness, efficiency and satisfaction. After some adjustment, there are four criteria important in web applications which are efficiency, errors, satisfaction and effectiveness. Beside, these three criteria also relate to tool testing. Table I provides a description of each criteria.

TABLE I. UAT criteria

| Criteria | Description |
|---|---|
| Efficiency | • Users' achieve certain goals and the resources expended in achieving them.<br>• Including task completion time and learning time.<br>• Users can find information more quickly. |
| Errors | • Minimum error rate can be done during running time |
| Satisfaction | • Users' comfort with and positive attitude toward the use of web application.<br>• Users' satisfaction can be measured by attitude rating scales.<br>• User finds the overall website acceptable.<br>• Website being more pleasant to navigate page visit or site exploration. |
| Effectiveness | • Users achieve certain goals.<br>• Including quality of solution and error rates.<br>• Goals for overall system are achieved. |

Based on the description of criteria in Table I, efficiency of web application can be tested by calculating task completion time or learning time. Task completion time is the calculated amount of time required for any particular task to be completed. Errors can be tested by calculating the minimum error rate done by user during running time while completing tasks. It is actually the number of error done by user when they try to enter some input or did not enter any input where the input is invalid. Satisfaction of web application relate to how friendly is that web page and satisfaction can be measured or tested by attitude rating scale where it is rating scale given by user after page navigation. Effectiveness of web application is that user able to accomplish task and achieves goal. This criterion can be tested by calculating time taken for user to finish one complete task as their goal.

There are two possible test cases for this evaluation phase. Figure 2 and Figure 3 are the sample of first test cases to test all the criteria. The first test case is about user successfully login and logout into system. Valid username and password required in entering the system. So, in testing efficiency and effectiveness, amount of time for user to completely login and logout into and from the system are calculated. User also has to achieve their goal. In this case, the user has to successfully enter the system. For errors in web application, number of errors is calculated during user enters username and password either they have entered valid username and password or vice versa.

The second test case is about user successfully updating their personal information into the system. Figure 4, Figure 5 and Figure 6 show step by step how user updates their profile. There are three elements in the form that cannot be empty and it is required. Same goes for next test case, in testing efficiency, effectiveness and satisfaction, amount of time for user to completely update their profile into the system are calculated. There are goals that should be achieved where user will successfully update their profile. For errors, it will be calculated during updating profile process.

Step No 1:

Step Description: Open URL elearning.utm.my?/14152/

Expected Result: Login page is displayed and contain the below fields

a) "Username" test field.
b) "Password" text field.
c) "Login" button.

Step No 2:

Step Description: Once Login page is displayed. Enter valid username/password.

a) Enter "wendy" in "Username" text field.
b) Enter "1234wendy" in "Password" text field.

Expected Result:

a) Verify "Username" is populated with text "wen".
b) Verify text entered in "Password" field is masked and is not readable.

Fig. 2. Example of test case to test criteria of web application

Step No 3:

Step Description: Click "Submit" button

Expected Result: Verify application "Home" page is displayed

a) Verify "Home" page displays "You are logged in as SHEROLWENDY ANAK SUALIM A12CS0120" message on top of right navigation Menu.
b) Verify right navigation menu contains links "Logout".

Step No 4:

Step Description: Click "Logout" link on the left menu.

Expected Result: Verify user is successfully logged out of the application and application login page is displayed elearning.utm.my?/14152/

Fig. 3. Example of test case to test criteria of web application (continued)

Step No 1:

Step Description: Update profile in Edit Profile page.

Expected Result: Edit profile page is displayed and contain the below fields.

a) "First name" text field.
b) "Surname" text field.
c) "Email address" text field.
d) "Email display" dropdown.
e) "City/town" text field.
f) "Select a country" dropdown.
g) "Timezone" dropdown.
h) "Description" text field.
i) "Update Profile" button.

Fig.4. Examples of test case to test criteria of web application (continued)

Step No 2:

Step Description: Once Edit Profile page is displayed. Enter valid information.

a) Enter "Natasha" in "First Name" text field.
b) Enter "Tasha" in "Surname" text field.
c) Enter "natasha@utm.my" in "Email address" text field.
d) Select "Hide my email address" for "Email display" dropdown.
e) Enter "Skudai" in "City/town" text field.
f) Select "Malaysia" for "Country" dropdown.
g) Select "Server timezone (Asia/Kuala_Lumpur)" time zone.
h) Enter "Student" in "Description" text field.

Expected Result:

a) Verify "First Name" is populated with text "Natasha" and not empty.
b) Verify "Surname" is populated with text "Tasha" and not empty.
c) Verify "Email Address" is in correct format.

Fig.5. Examples of test case to test criteria of web application (continued)

Step No 3:

Step Description: Click "Update Profile" button.

Expected Result: Verify user is successfully updated profile and "Preferences" page is display.

Fig.6. Examples of test case to test criteria of web application (continued)

There are justifications on choosing test cases for testing criteria of web application. Based on the test cases, login and logout process and also fill up form are two important phases in web application. Besides, there will be errors when inserting data for both phases either during login or update details in the web application. So, these four criteria of web application will be fully tested.

### *B. Set of Questions for Evaluation*

Every tool will be evaluated according to the set of questions assigned for each criteria and general criteria as shown in Table II. Some of the questions are tested in the prototype and some of them are tested based on observation.

TABLE II. EVALUATION QUESTIONS

| Criteria | ID | Questions | Possible Answers |
|---|---|---|---|
| General | Q1 | Level of easiness when using automated testing tool | • Easy<br>• Moderate<br>• Difficult |
| | Q2 | Can automated testing tool be used before or after the code implementation? | • Easy to implement<br>• Difficult to implement<br>• Not possible |
| | Q3 | Can automated testing tool be applied for web application? | • Applied for web application<br>• Applied for non-web application |

*(…cont.)*

| | | | |
|---|---|---|---|
| | Q4 | How to create test cases for GUI tests? | • Coding<br>• Capture and Replay |
| | Q5 | Does the automated testing tool use id"s or position coordinates of GUI elements? | • Id-based representation<br>• Position-based representation |
| | Q6 | Can the automated testing tool access and check error messages? | • Possible<br>• Not possible |
| | Q7 | Can the automated testing tool interact with database? | • Possible<br>• Not possible |
| Efficiency | Q8 | How good the degree of accuracy and completeness web based application can provide information and functionality according to :<br>1. Quality of solution for errors<br>2. Mistake recovery | • Efficient<br>• Moderate<br>• Not al all |
| Effectiveness | Q9 | How effective that web application based on :<br>1. Logical partition of information<br>2. Proper link text<br>3. Loading of proper style sheets<br>4. Message and dynamic data management<br>5.Terminological consistency and layout | • Effective<br>• Moderate<br>• Not at all |
| Satisfaction | Q10 | What is degree of difficulty users can find important information in web application based on :<br>1. Number of link and frames<br>2. Proper name of the frames<br>3. Location of navigation link | • Easy<br>• Moderate<br>• Not at all |
| | Q11 | What is degree of difficulty users can find important information in web application based on:<br>1.Number of link and frames<br>2.Proper name of the frames<br>3.Location of navigation link | • Easy<br>• Moderate<br>• Not at all |

*(…cont.)*

| | | | |
|---|---|---|---|
| | Q13 | How pleasant that web application to navigation of page visit or site exploration based on :<br>1. Addition of short sound<br>2. Color of text and background<br>3. Magnifying at passing by mouse pointer | • Pleasant<br>• Moderate<br>• Hard |
| Errors | Q12 | How maximum is error rate can be detected? | • High<br>• Low |

## III. Results and Discussion

Table III shows the findings of the evaluation of the three tools. The comparison is made due to the last objective which is providing the findings of the comparative study on automated acceptance testing tool for web application. There are relevant answers from other thesis writing that support the results. The important of the findings are to help user compares and choose suitable tool for testing as well as providing elements and aspect should be checked.

TABLE III. Tools Evaluation

| ID | Tools | | |
|---|---|---|---|
| | **TestComplete** | **Selenium Webdriver** | **Watir Webdriver** |
| Q1 | Easy | Moderate | Moderate |
| Q2 | Easy | Easy | Easy |
| Q3 | Applied for web application | Applied for web application | Applied for web application |
| Q4 | Capture and replay<br>Coding | Coding | Coding |
| Q5 | Position-based coordinates | Position-based and id-based coordinate | Id-based coordinates |
| Q6 | Possible | Possible | Possible |
| Q7 | Possible | Possible | Possible |
| Q8 | Moderate | Efficient | Efficient |
| Q9 | Moderate<br>**Execution time:**<br>**56.89 seconds**<br>**4.16 minutes** | Effective<br>**Execution time:**<br>**9.47 seconds**<br>**3.20 minutes** | Effective<br>**Execution time:**<br>**9.2 seconds**<br>**3.05 minutes** |
| Q10 | Moderate<br>**Execution time:**<br>**1.29 minutes**<br>**5.05 minutes** | Easy<br>**Execution time:**<br>**30.16 seconds**<br>**3.55 minutes** | Easy<br>**Execution time:**<br>**16.1 seconds**<br>**3.49 minutes** |
| Q11 | Moderate<br>**Execution time:**<br>**1.29 minutes**<br>**5.05 minutes** | Pleasant<br>**Execution time:**<br>**30.16 seconds**<br>**3.55 minutes** | Pleasant<br>**Execution time:**<br>**16.1 seconds**<br>**3.49 minutes** |
| Q12 | High | High | High |
| Q13 | Moderate<br>**Execution time:**<br>**1.29 minutes**<br>**5.05 minutes** | Pleasant<br>**Execution time:**<br>**30.16 seconds**<br>**3.55 minutes** | Pleasant<br>**Execution time:**<br>**16.1 seconds**<br>**3.49 minutes** |

The findings show that TestComplete tool ease tester with the ability to create, manage and run tests from any Windows or Web, making it easy to quickly create and execute tests with efficient playback. Both Selenium and Watir webdrivers tools are moderate to use as experience is compulsory due to the requirement of programming skills in ruby. Without ruby knowledge, it is not possible to write the script for web application. All testing tools support testing desktop, web and mobile applications. These three testing tools use coding technique to create test cases. Coding technique drives the design to be flexible and maintainable direction as well as changing certain functions without any affection. However, during implementation TestComplete use capture and replay where record-play system has to rerecord again if there is any aspect of page change. This makes the test case readable for non-software engineer. The used of position coordinate during recording script in TestComplete and Selenium Webdriver are reliable enough. When TestComplete record coordinates, it uses relative coordinates because the size of button is measured in pixels and it does not change on different resolution. If the button is moved to another position on the form, the script will still work. When executing the test command, the first thing checked by TestComplete is whether the object exists in the system and then executes the needed method or property of the object. Selenium Webdriver mostly using position coordinate in test script and sometimes id coordinates when needed. The unique part about Selenium test script is that it will check first the object if exist and then execute the command. Differ from Watir Webdriver where it is using id coordinates during implementation. Test script for Watir Webdriver is simple compares to Selenium Webdriver because it directly accesses the object using id and trigger error later. That is why the test script looks simple but it actually specific. As both Selenium and Watir webdrivers are using ruby language, their test cases are hardly readable for non-software engineer. All of the three testing tools are able to check error message and also possible to interact with database. TestComplete supports working with database and allows both reading and writing it to database. Ruby gives the user the power to connect to databases.

The running time for automated testing tools was recorded ten times and the average is calculated. The average runtime is actually the duration for automated testing tool to complete a task given. The performance level of the testing tool will be determined by the time taken to complete the task. Degree of accuracy and completeness of web application for TestComplete is moderate as this testing tool takes extra time for error solution and mistake recovery. But for Selenium and Watir webdrivers, both are very effective as they take shorter time to overcome mistake and error. For effectiveness and difficulty of web application, TestComplete shows the moderate results in order to read and navigate link and also to load page using coordinates. Selenium and Watir show the effective results for effectiveness and difficulty of web application as they also take shorter time to read and navigate pages and load pages using id and position coordinates. TestComplete, Selenium Webdriver and Watir Webdriver are testing tools that can detect high maximum of error rate. Every error detected by TestComplete tool recorded in test log. Even single errors that have been detected will pop up at the end before testing finish. The web application form is more pleasant as all the testing tools able to navigate pages based on color of text and background.

TestComplete can be concluded as a user friendly testing tool that can be used to test simple web application. TestComplete has easy to use UI and efficient playback. TestComplete will be best to use for application with lesser security needs. Thus it does not provide data security while testing as it is using position-based coordinates. Selenium Webdriver makes direct call to the browser using browser's built in support for automation. There is no server needed for Selenium Webdriver. With the simple test script and the use of position and id coordinates helps this testing tool to be more suitable for web application testing. The answers for all web application questions also mostly positives. Watir Webdriver is as good as Selenium Webdriver. But there is slightly different in test script where Watir Webdriver is using fully id coordinate with the absence of xpath selectors in order to make testing more accurate. Watir Webdriver is suitable under certain specific situation compare to Selenium Webdriver.

## IV. Conclusion

Usability criteria such as efficiency, effectiveness, errors and satisfaction are important criteria in evaluating the user acceptance of a tool. The difference in execution time will differentiate and classify either the best testing tool or not. Based on the evaluation performed in this research, Watir Webdriver is the chosen automated testing tool for web application among the three evaluated tools. The tool is not only has the shortest execution time and excellent test script but also fulfill usability criteria as well. Watir Webdriver shows the efficiency criteria when it has minimum rate of speed to complete task given. Besides, Watir Webdriver also has error criteria with high maximum error rate detected during testing web application. This testing tool also has high satisfaction as it is pleasant to use the tool during testing web application. As a conclusion, with that usability criteria checked and other special features Watir Webdriver is the best choice for tester to perform testing for web application.

## Acknowledgment

We would like to thank Ministry of Higher Education Malaysia for sponsoring the research through the Fundamental grant "Search-Based Mobile Applications Test Data Generation Model" and Universiti Teknologi Malaysia for providing the facilities and support for the research.

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